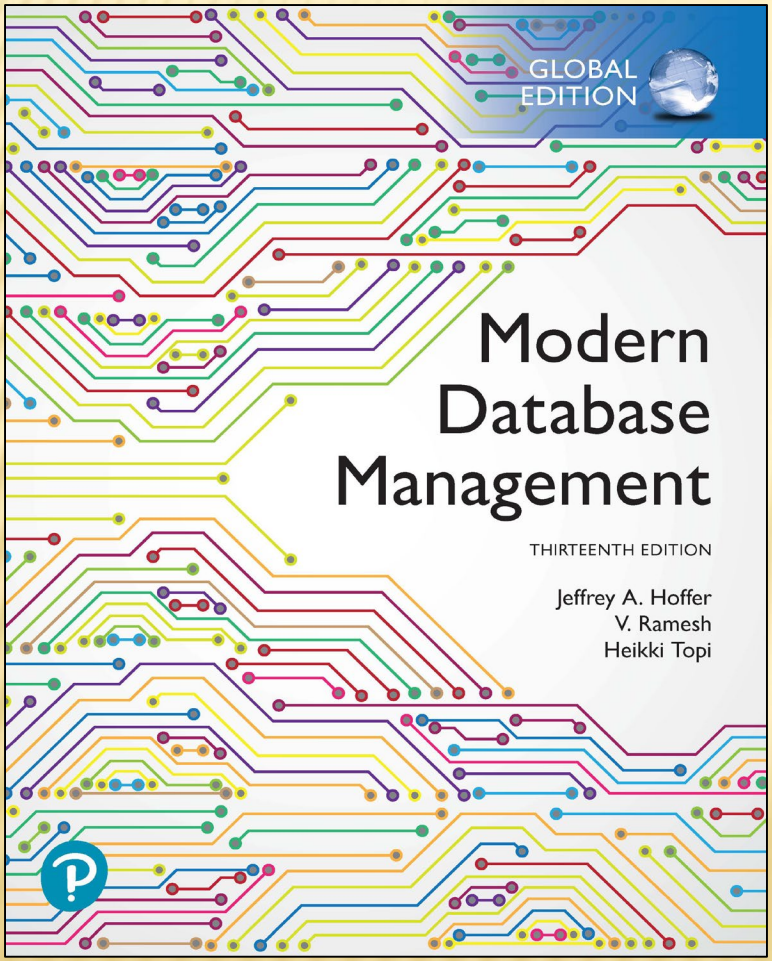
Modern Database Management

Thirteenth Edition



Chapter 3

The Enhanced E-R Model

LEARNING OBJECTIVES

3.1 Define terms

3.2 Understand use of supertype/subtype relationships

3.3 Use specialization and generalization techniques

3.4 Specify completeness and disjointness constraints

3.5 Develop supertype/subtype hierarchies for business situations

3.6 Develop entity clusters

3.7 Explain universal (packaged) data model

3.8 Describe special features of data modeling project using packaged data model

SUPERTYPES AND SUBTYPES

- **Enhanced E-R (EER) model**: extends original E-R model with new modeling constructs
- **Subtype**: A subgrouping of the entities in an entity type that has attributes distinct from those in other subgroupings
- **Supertype**: A generic entity type that has a relationship with one or more subtypes
- **Attribute Inheritance**:
  - Subtype entities inherit values of all attributes and relationships of the supertype
  - An instance of a subtype is also an instance of the supertype

FIGURE 3-1 BASIC NOTATION FOR SUPERTYPE/SUBTYPE NOTATION (1 OF 2)

a) EER notation

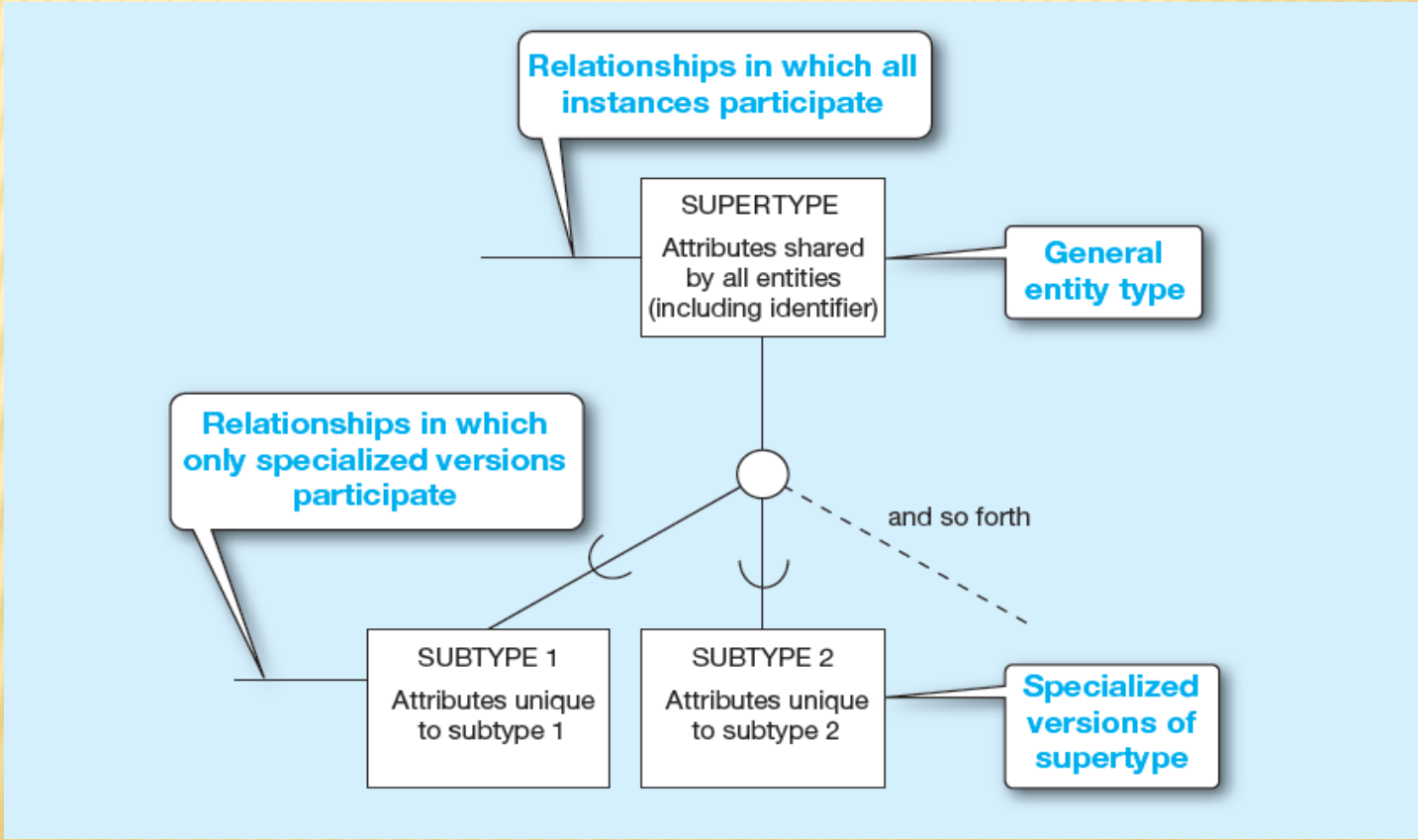


FIGURE 3-1 BASIC NOTATION FOR SUPERTYPE/SUBTYPE NOTATION (2 OF 2)

b) Microsoft Visio Notation

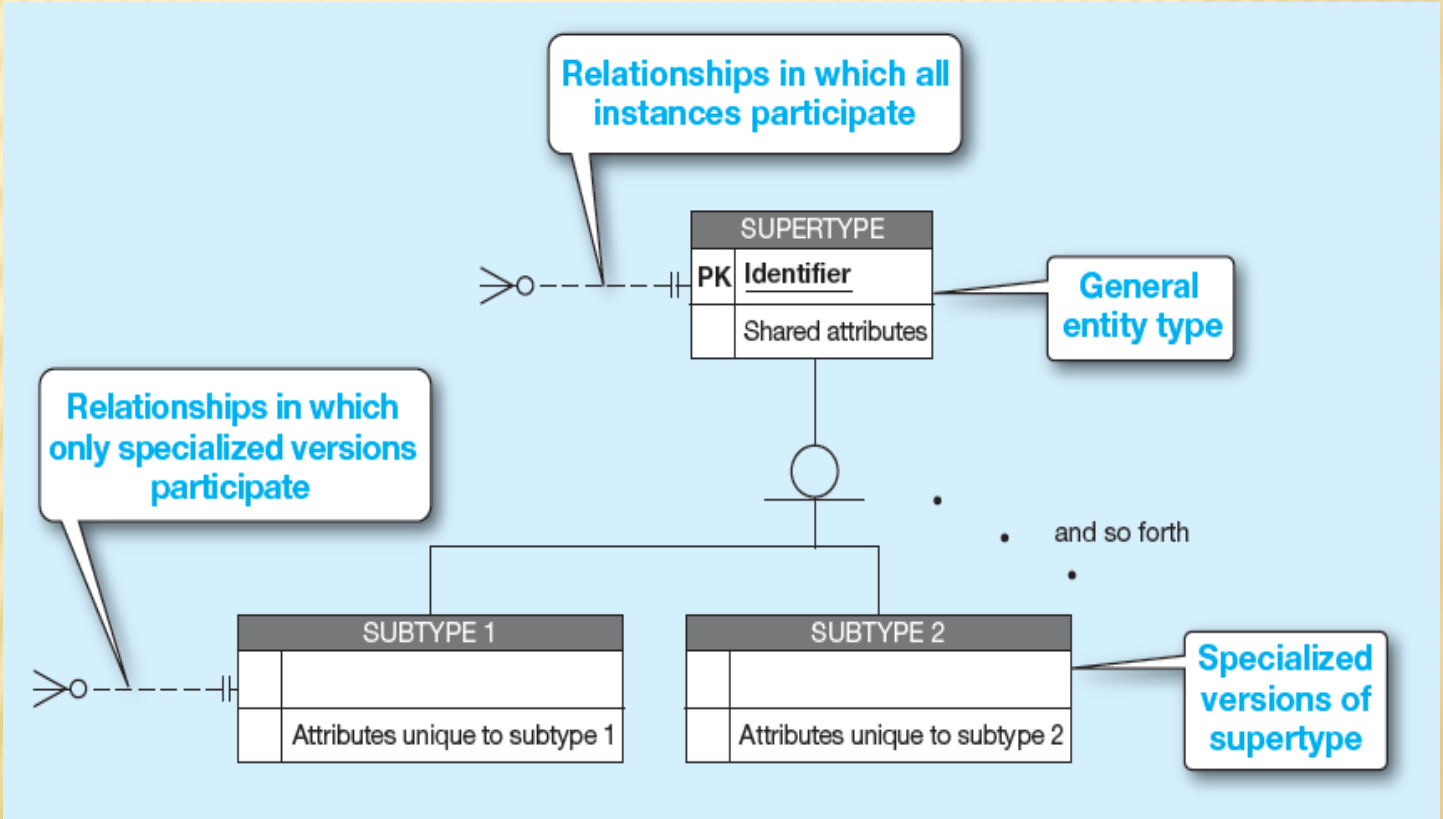
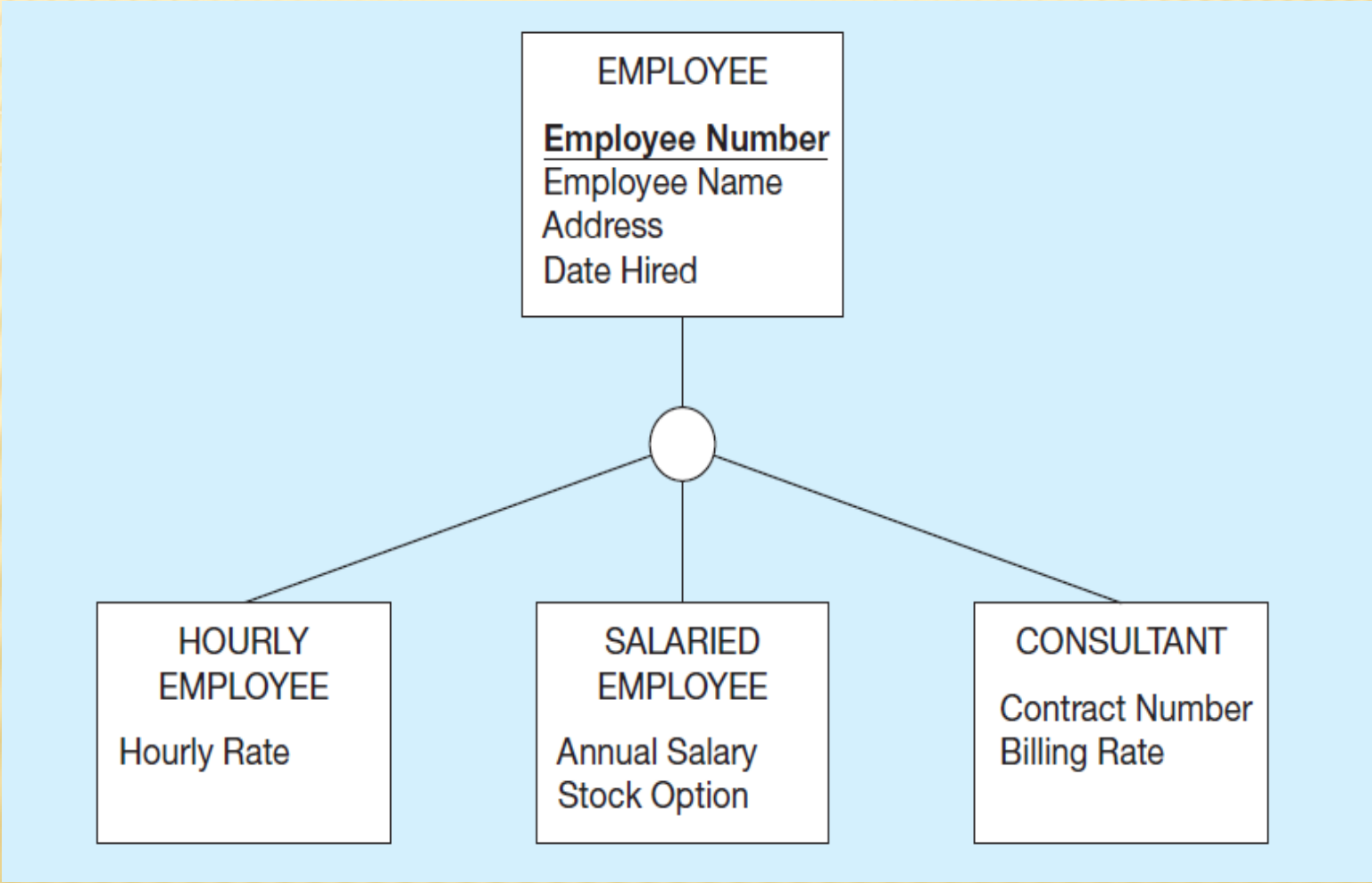


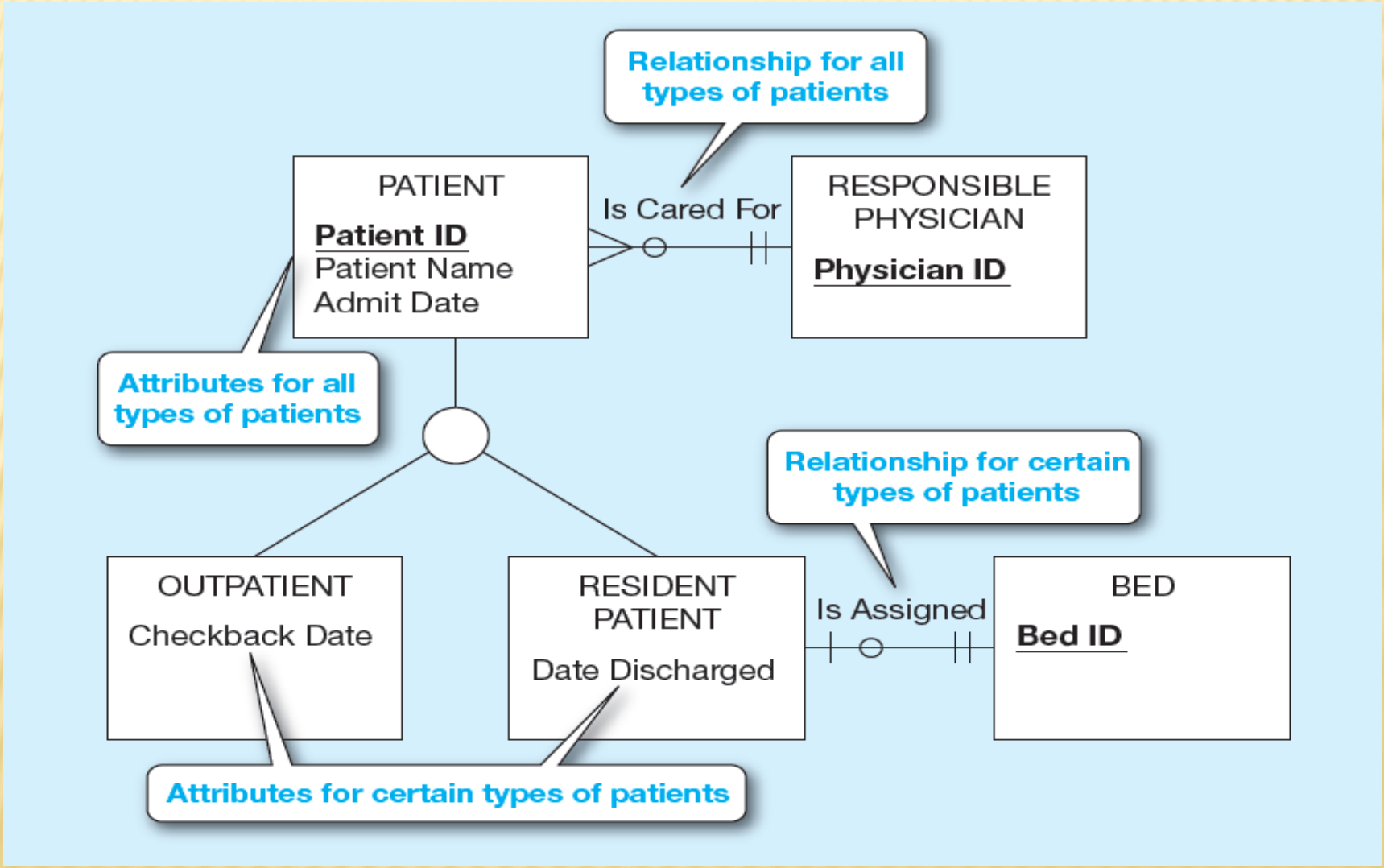
FIGURE 3-2 EMPLOYEE SUPERTYPE WITH THREE SUBTYPES



RELATIONSHIPS AND SUBTYPES

- Relationships at the **supertype** level indicate that all subtypes will participate in the relationship
- The instances of a **subtype** may participate in a relationship unique to that subtype. In this situation, the relationship is shown at the subtype level

Figure 3-3 Supertype/Subtype Relationships in a Hospital



GENERALIZATION AND SPECIALIZATION

- **Generalization**: The process of defining a more general entity type from a set of more specialized entity types. BOTTOM-UP
- **Specialization**: The process of defining one or more subtypes of the supertype and forming supertype/subtype relationships. TOP-DOWN

FIGURE 3-4 EXAMPLE OF GENERALIZATION (1 OF 2)

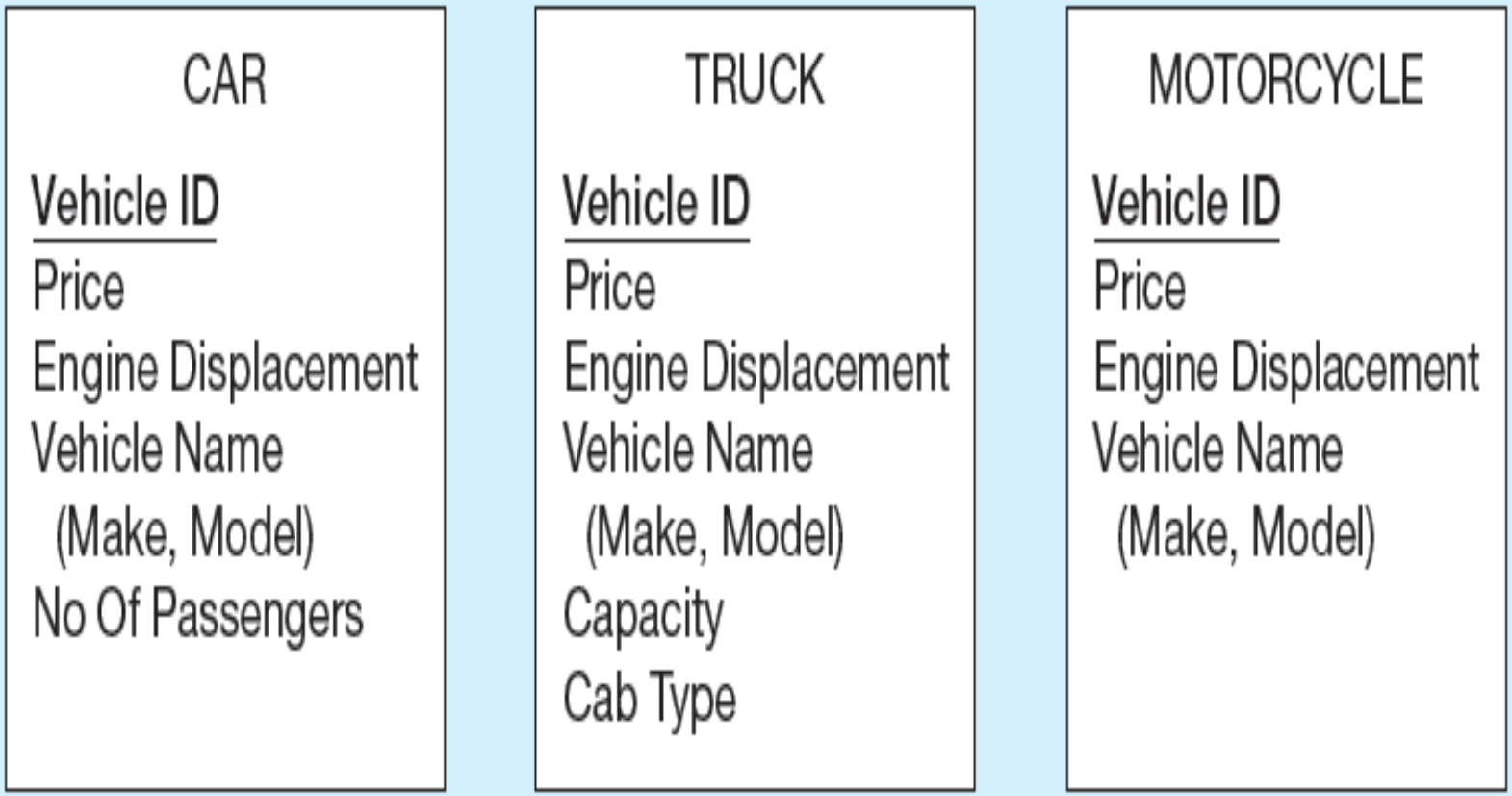


FIGURE 3-4 EXAMPLE OF GENERALIZATION (2 OF 2)

b) Generalization to VEHICLE supertype

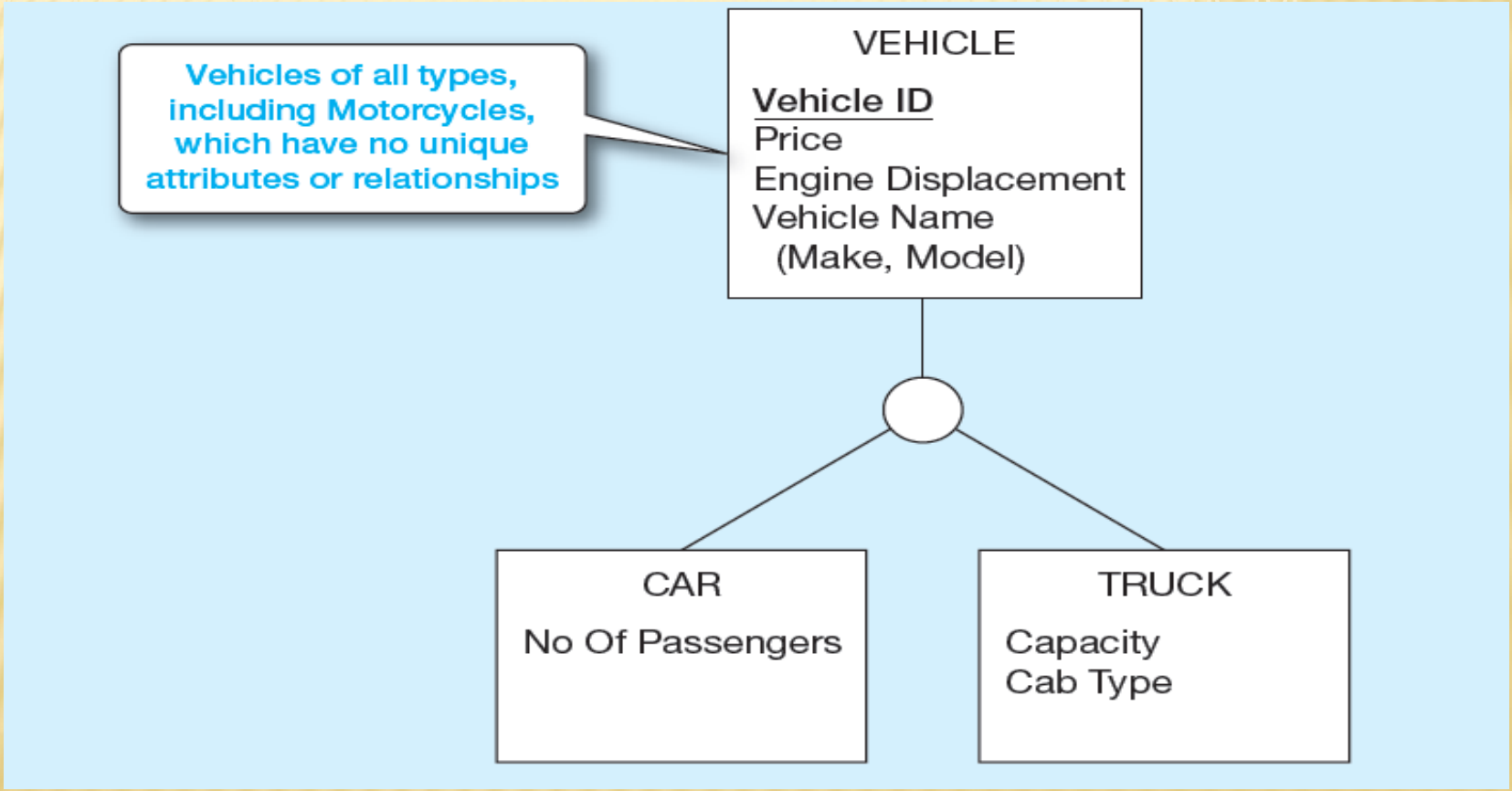


FIGURE 3-5 EXAMPLE OF SPECIALIZATION (1 OF 2)

a) Entity type PART

Note: For this part, the manufacturer decided that it should be produced in house.

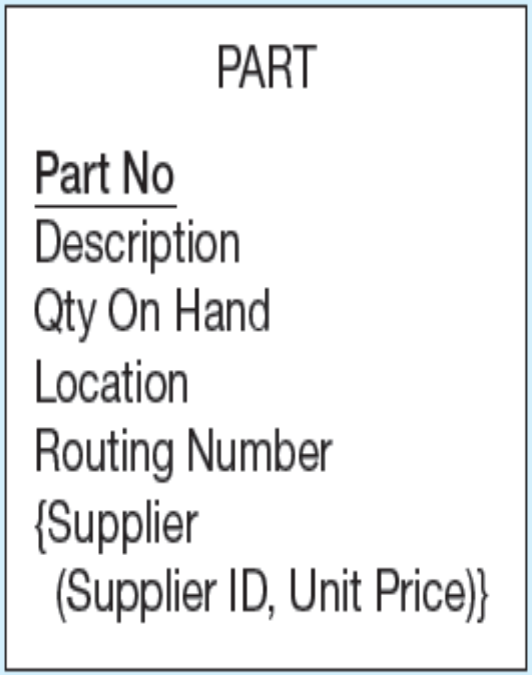
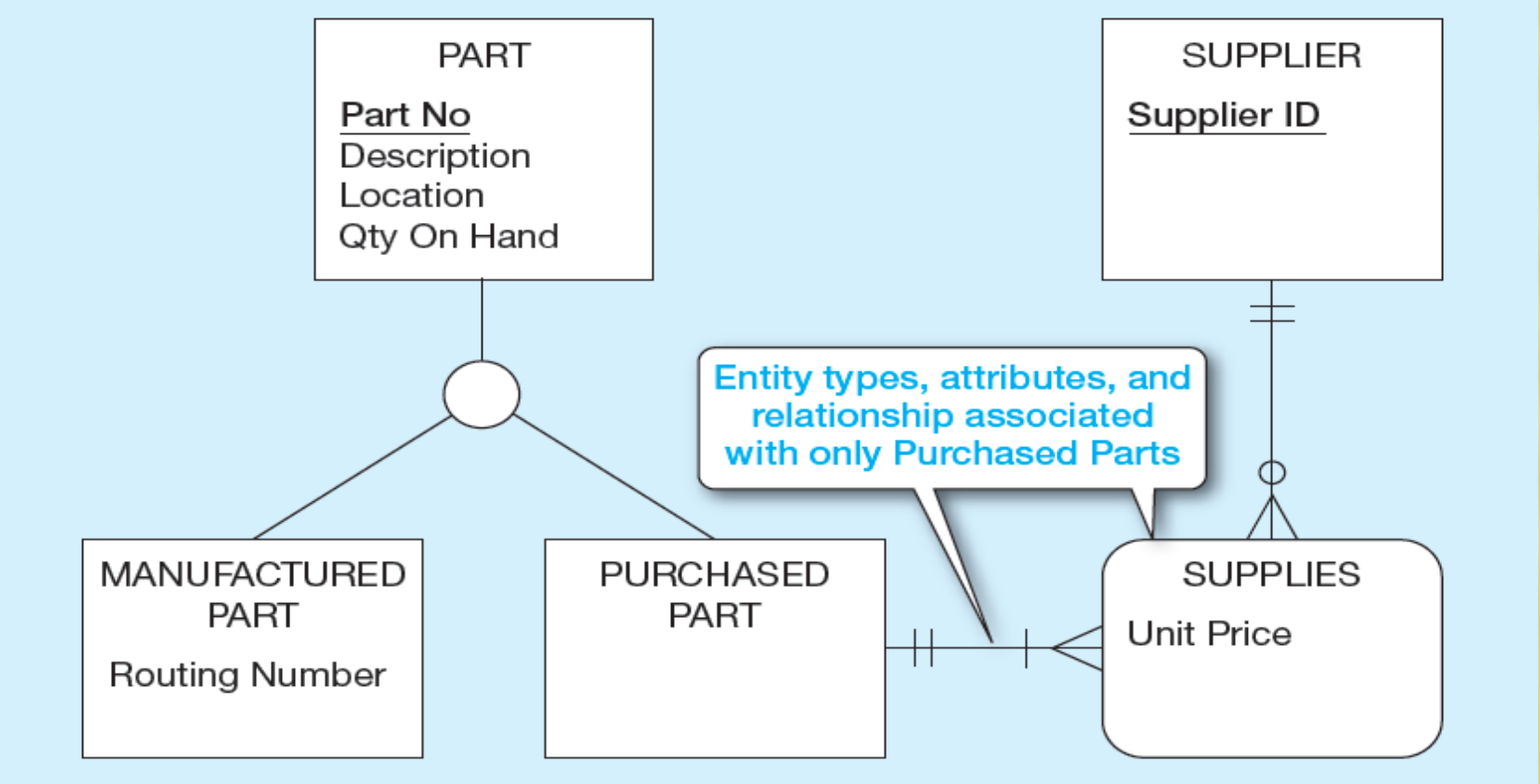


FIGURE 3-5 EXAMPLE OF SPECIALIZATION (2 OF 2)

b) Specialization to MANUFACTURED PART and PURCHASED PART

Multivalued composite attribute replaced by associative entity relationship to another entity



CONSTRAINTS IN SUPERTYPE/SUBTYPE RELATIONSHIPS

- **Completeness Constraints**: Whether an instance of a supertype **must** also be a member of at least one subtype
  - Total Specialization Rule: Yes (double line)
  - Partial Specialization Rule: No (single line)

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FIGURE 3-6 EXAMPLES OF COMPLETENESS CONSTRAINTS (1 OF 2)

a) Total specialization rule

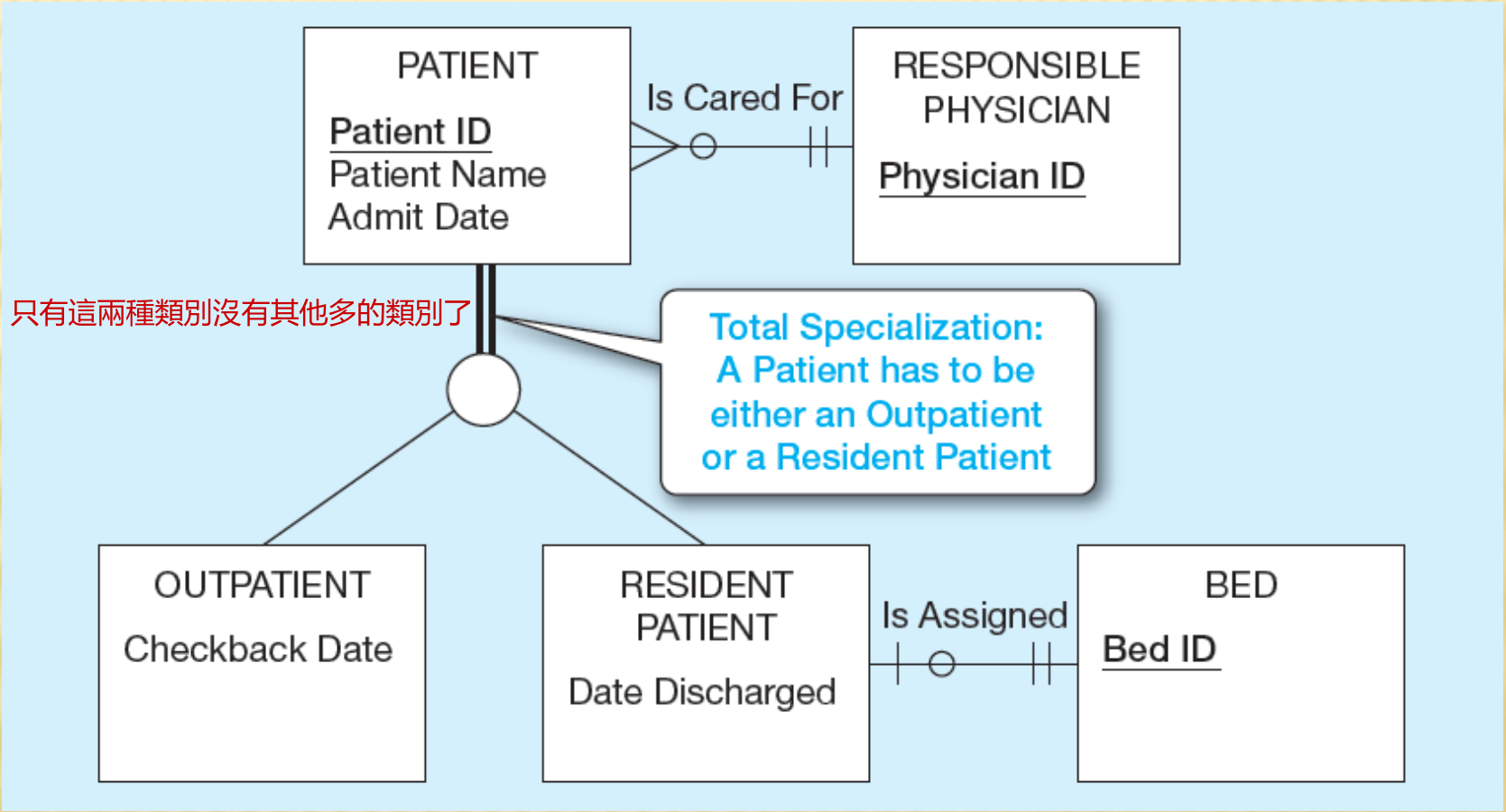
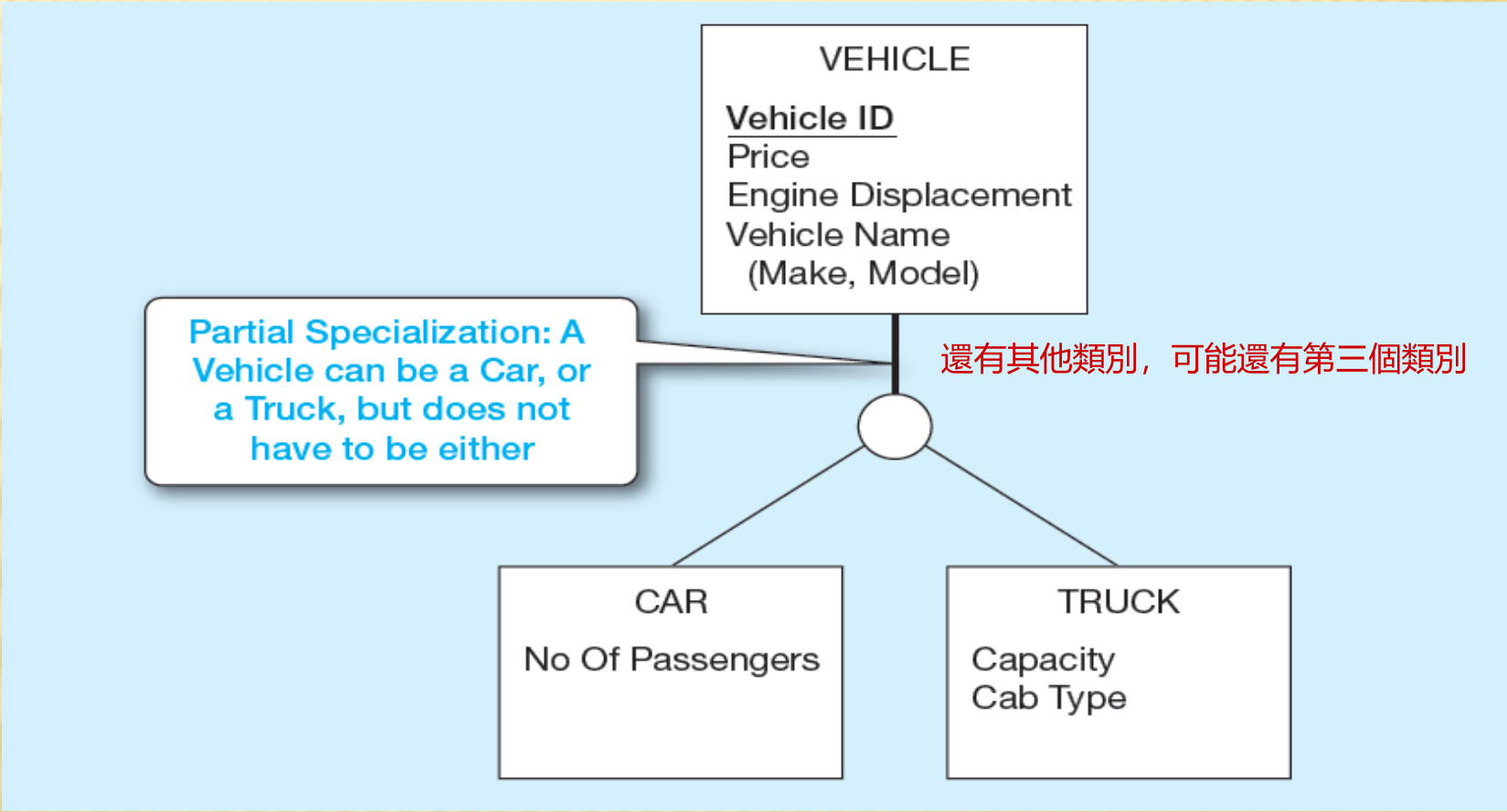


FIGURE 3-6 EXAMPLES OF COMPLETENESS CONSTRAINTS (2 OF 2)

b) Partial specialization rule



CONSTRAINTS IN SUPERTYPE/SUBTYPE RELATIONSHIPS (1 OF 2)

- **Disjointness Constraints**: Whether an instance of a supertype may **simultaneously** be a member of two (or more) subtypes
  - Disjoint Rule: An instance of the supertype can be only ONE of the subtypes
  - Overlap Rule: An instance of the supertype could be more than one of the subtypes

FIGURE 3-7 EXAMPLES OF DISJOINTNESS CONSTRAINTS (1 OF 2)

a) Disjoint rule

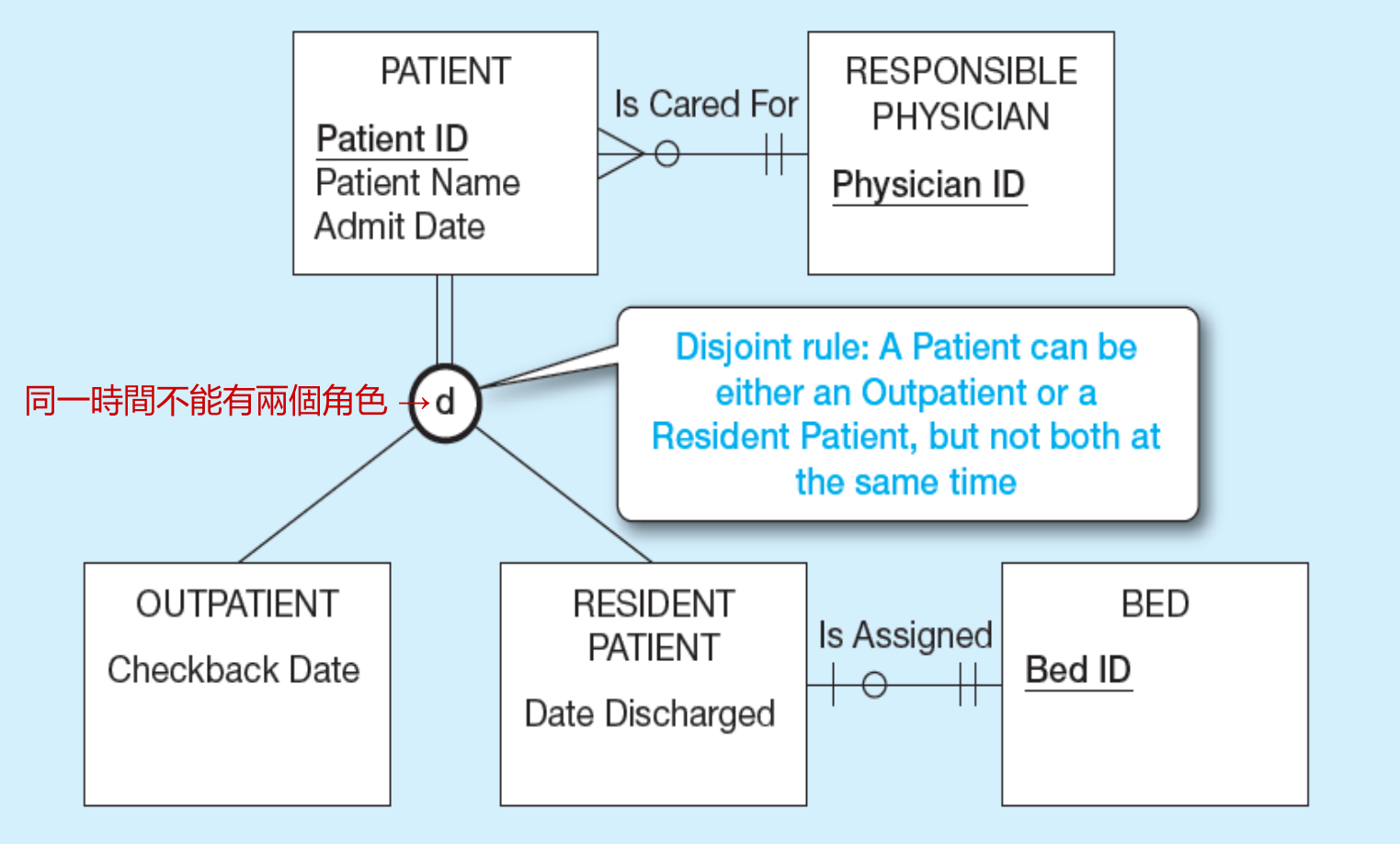
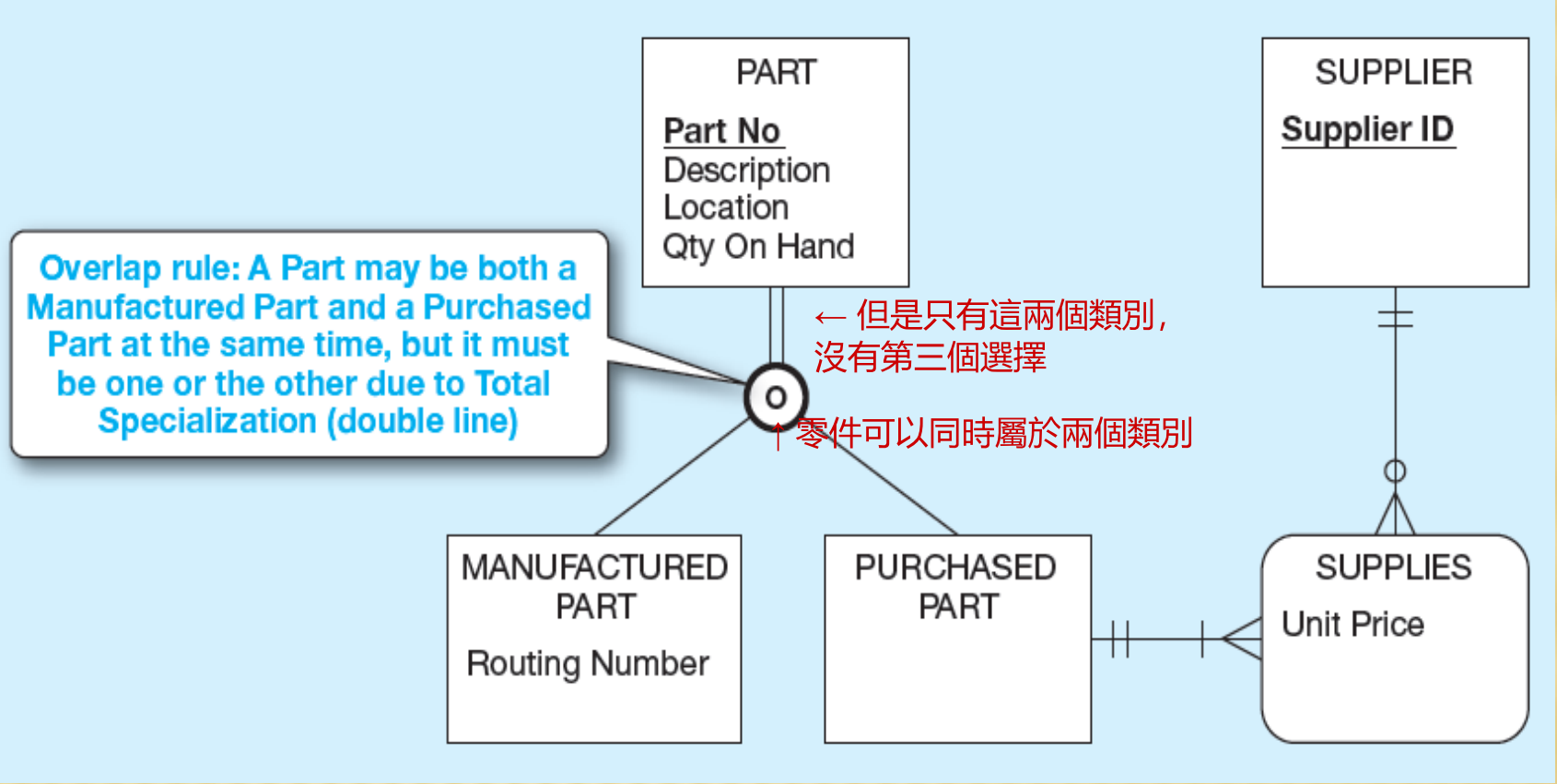


FIGURE 3-7 EXAMPLES OF DISJOINTNESS CONSTRAINTS (2 OF 2)

b) Overlap rule



CONSTRAINTS IN SUPERTYPE/SUBTYPE RELATIONSHIPS (2 OF 2)

- **Subtype Discriminator**: An attribute of the supertype whose values determine the target subtype(s)
  - **Disjoint** – a **simple** attribute with alternative values to indicate the possible subtypes
  - **Overlapping** – a **composite** attribute whose subparts pertain to different subtypes. Each subpart contains a Boolean value to indicate whether or not the instance belongs to the associated subtype

Figure 3-8 Introducing a Subtype Discriminator (Disjoint Rule)

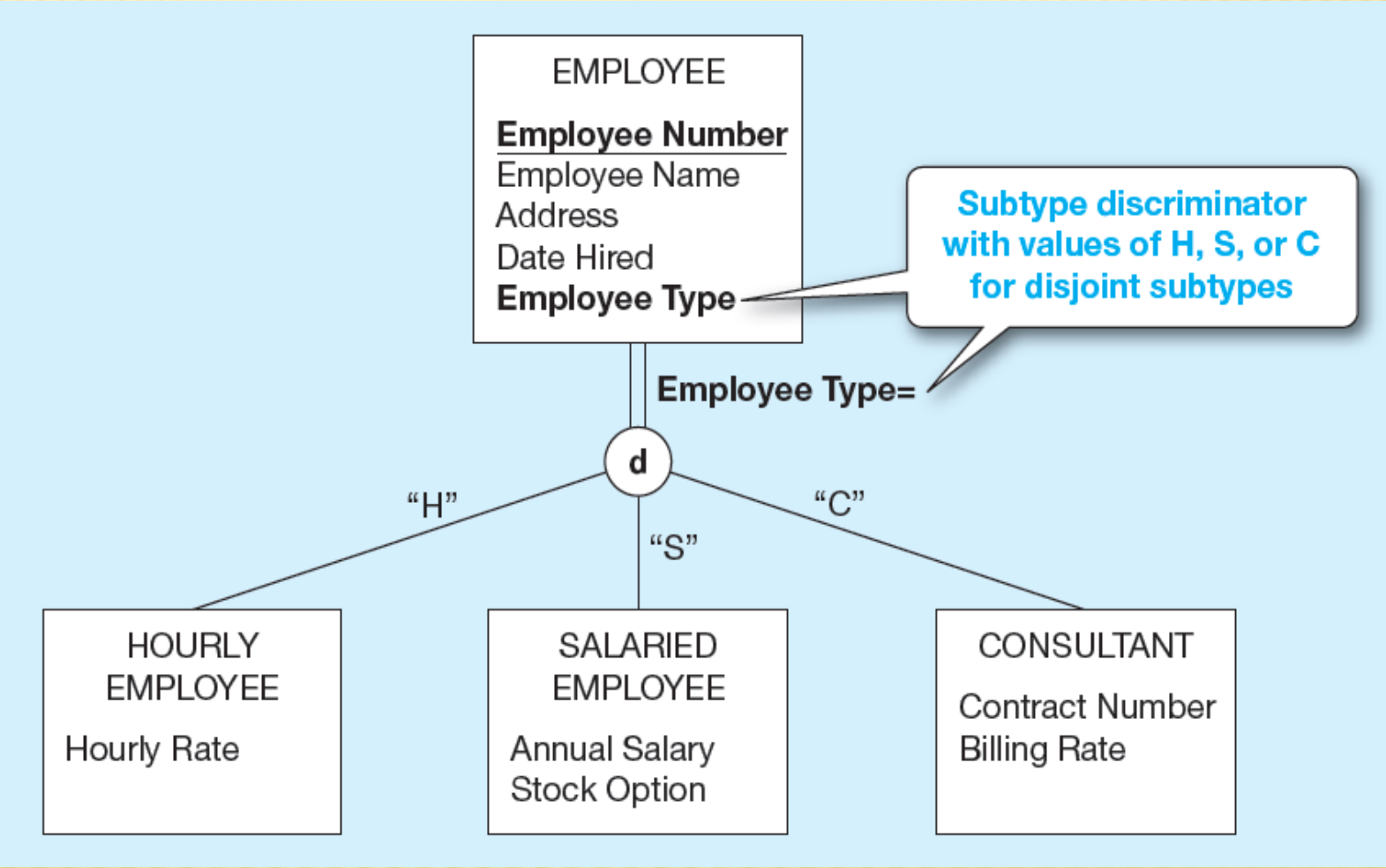


Figure 3-9 Subtype Discriminator (Overlap Rule)

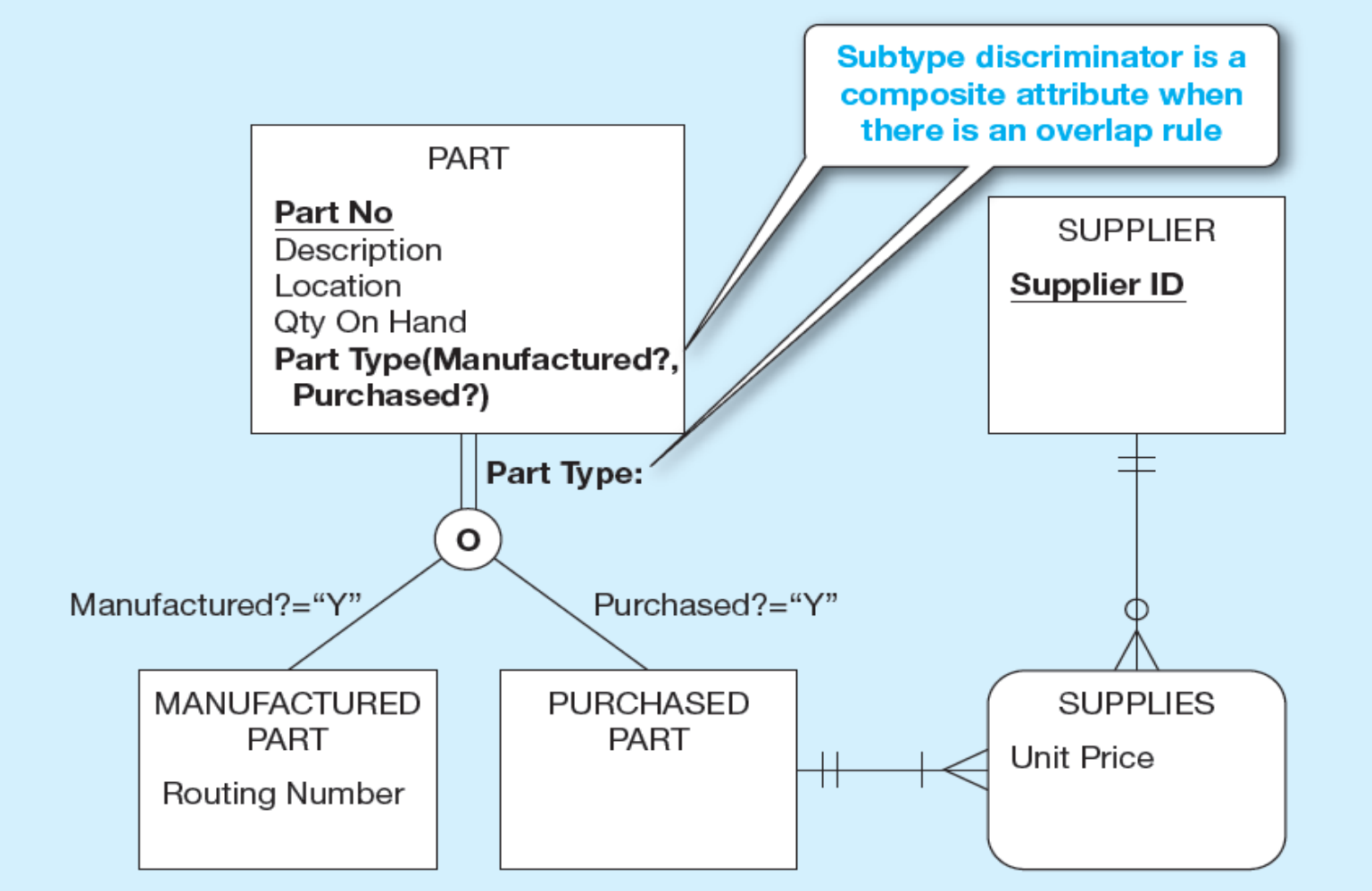
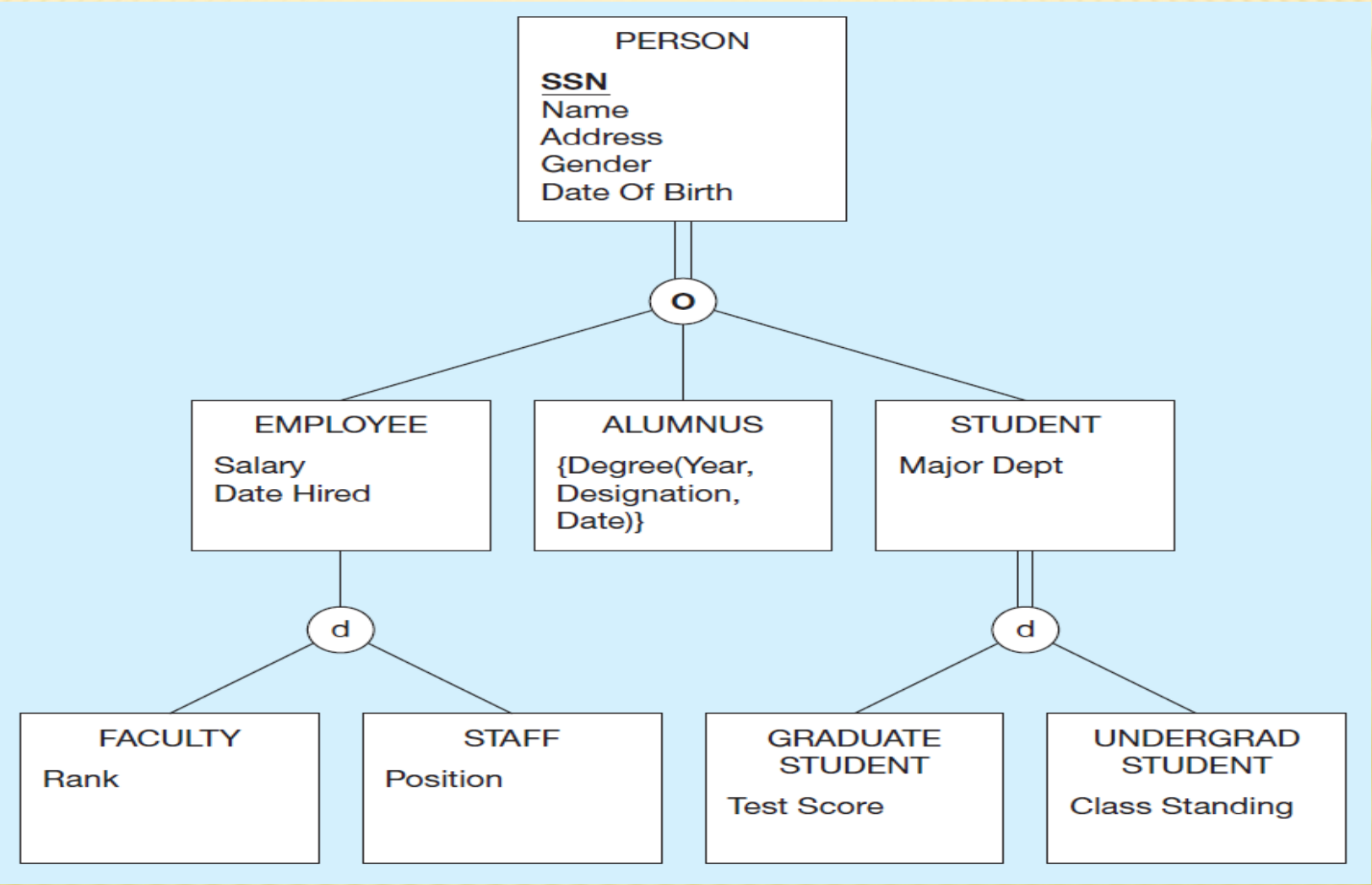


Figure 3-10 Example of Supertype/Subtype Hierarchy



ENTITY CLUSTERS 不考

- EER diagrams are difficult to read when there are too many entities and relationships.
- Solution: Group entities and relationships into **entity clusters.**
- **Entity cluster**: Set of one or more entity types and associated relationships grouped into a single abstract entity type

FIGURE 3-13 ENTITY CLUSTERING FOR PINE VALLEY FURNITURE COMPANY (1 OF 2)

a) Possible entity clusters (using Microsoft Visio)

Related groups of entities could become clusters

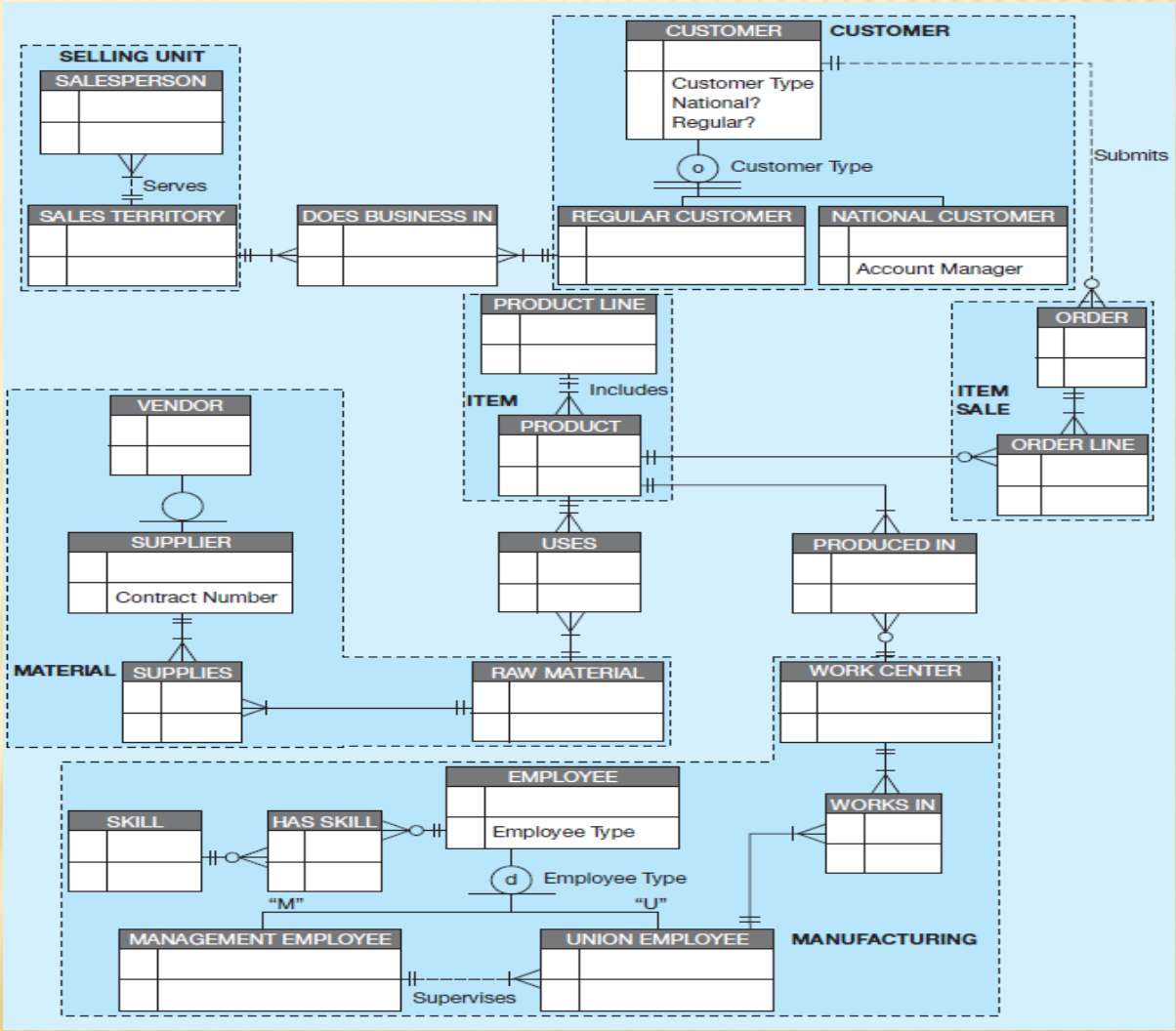


FIGURE 3-13 ENTITY CLUSTERING FOR PINE VALLEY FURNITURE COMPANY (2 OF 2)

b) EER diagram for entity clusters (using Microsoft Visio)

More readable, isn't it?

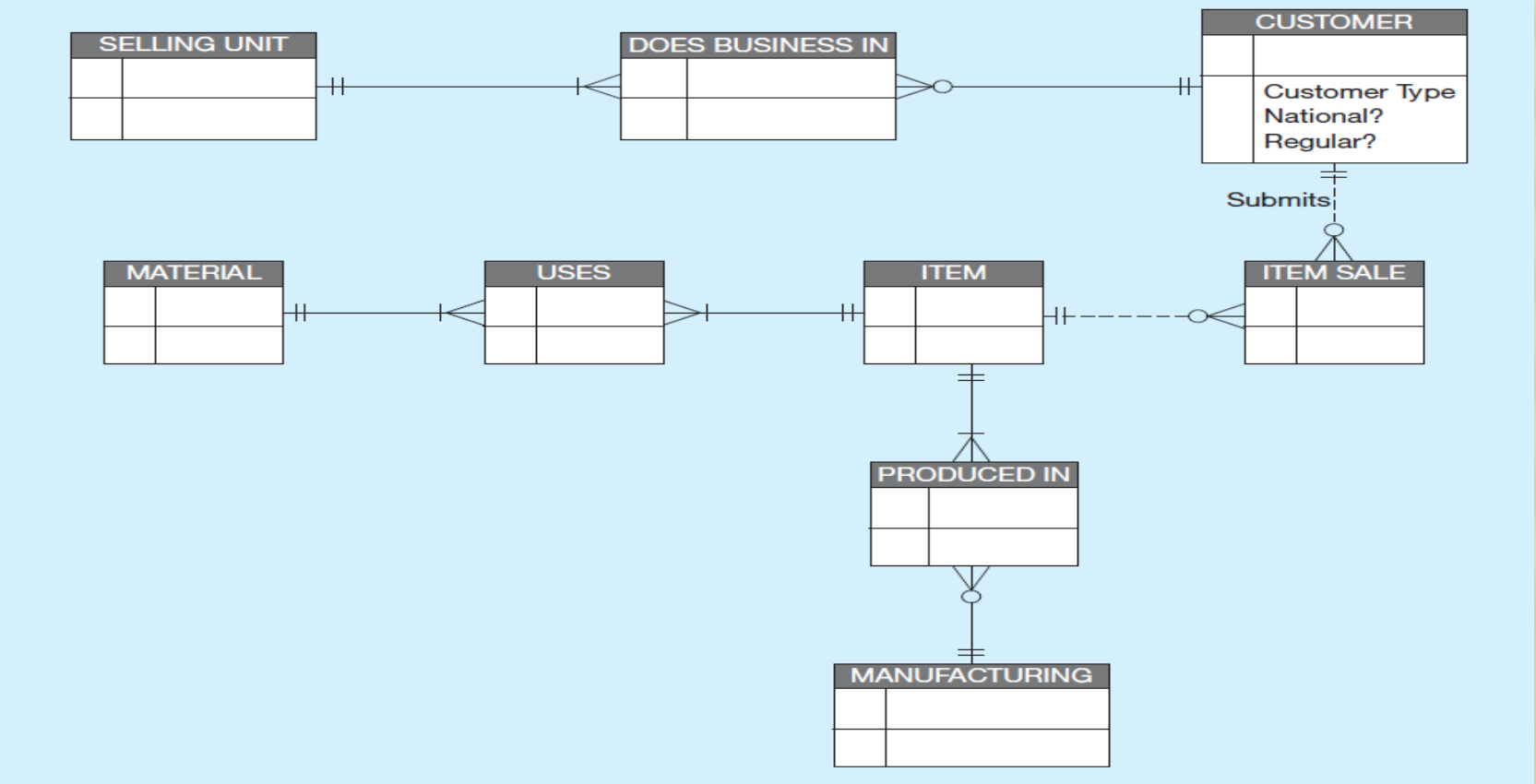
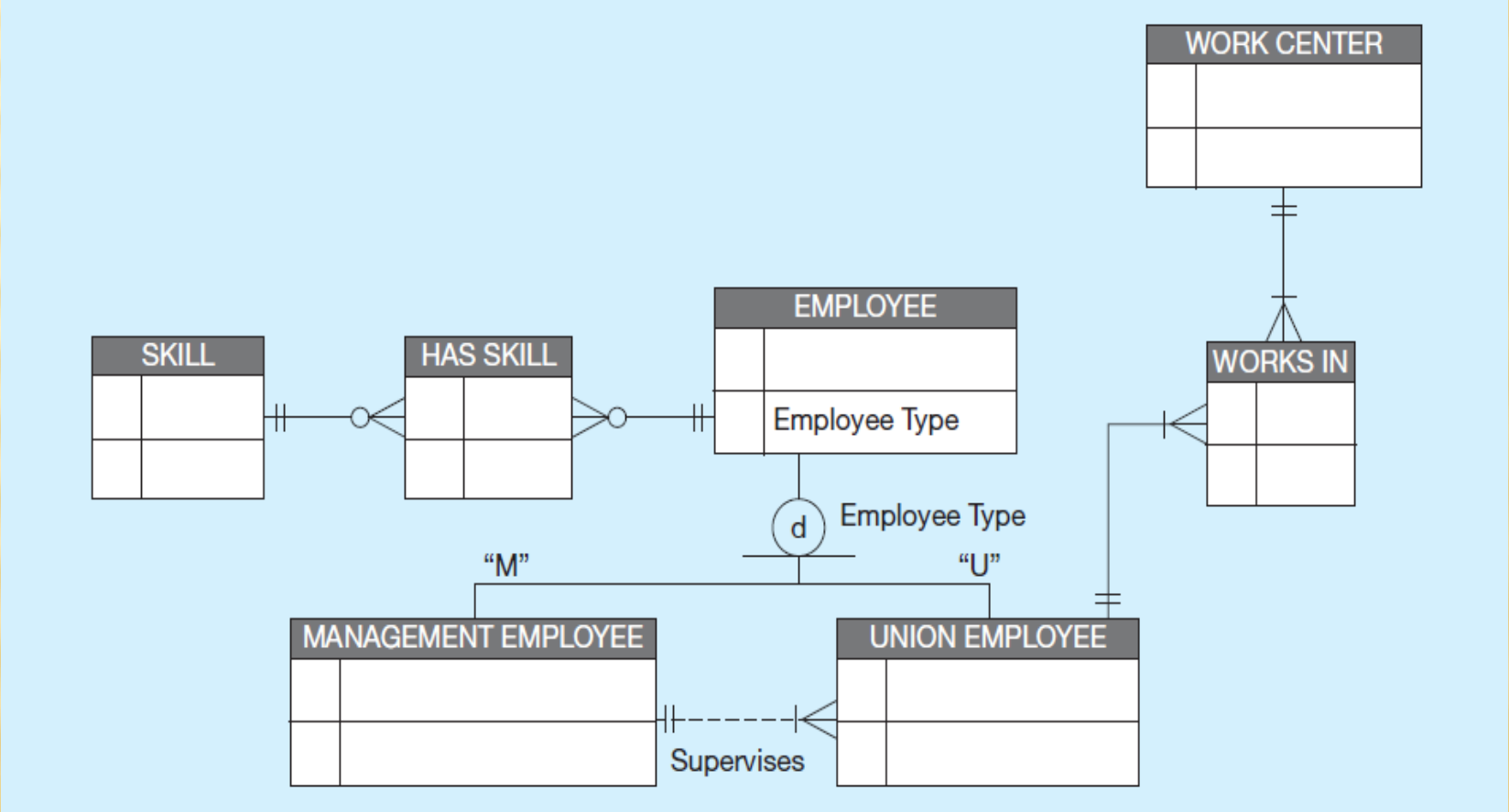


FIGURE 3-14 ENTITY CLUSTERING FOR PINE VALLEY FURNITURE COMPANY

Detail for a single cluster



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PACKAGED DATA MODELS

- Predefined data models
- Could be universal or industry-specific
- Universal data model = a generic or template data model that can be reused as a starting point for a data modeling project (also called a “pattern”)

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ADVANTAGES OF PACKAGED DATA MODELS (1 OF 2)

- Use proven model components
- Save time and cost
- Less likelihood of data model errors
- Easier to evolve and modify over time
- Aid in requirements determination
- Easier to read
- Supertype/subtype hierarchies promote reuse

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ADVANTAGES OF PACKAGED DATA MODELS (2 OF 2)

- Many-to-many relationships enhance model flexibility
- Vendor-supplied data model fosters integration with vendor's applications
- Universal models support interorganizational systems

FIGURE 3-15 PARTY, PARTY ROLE, AND ROLE TYPE IN A UNIVERSAL DATA MODEL (1 OF 2)

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a) Basic PARTY universal data model

Packaged data models are generic models that can be customized for a particular organization's business rules.

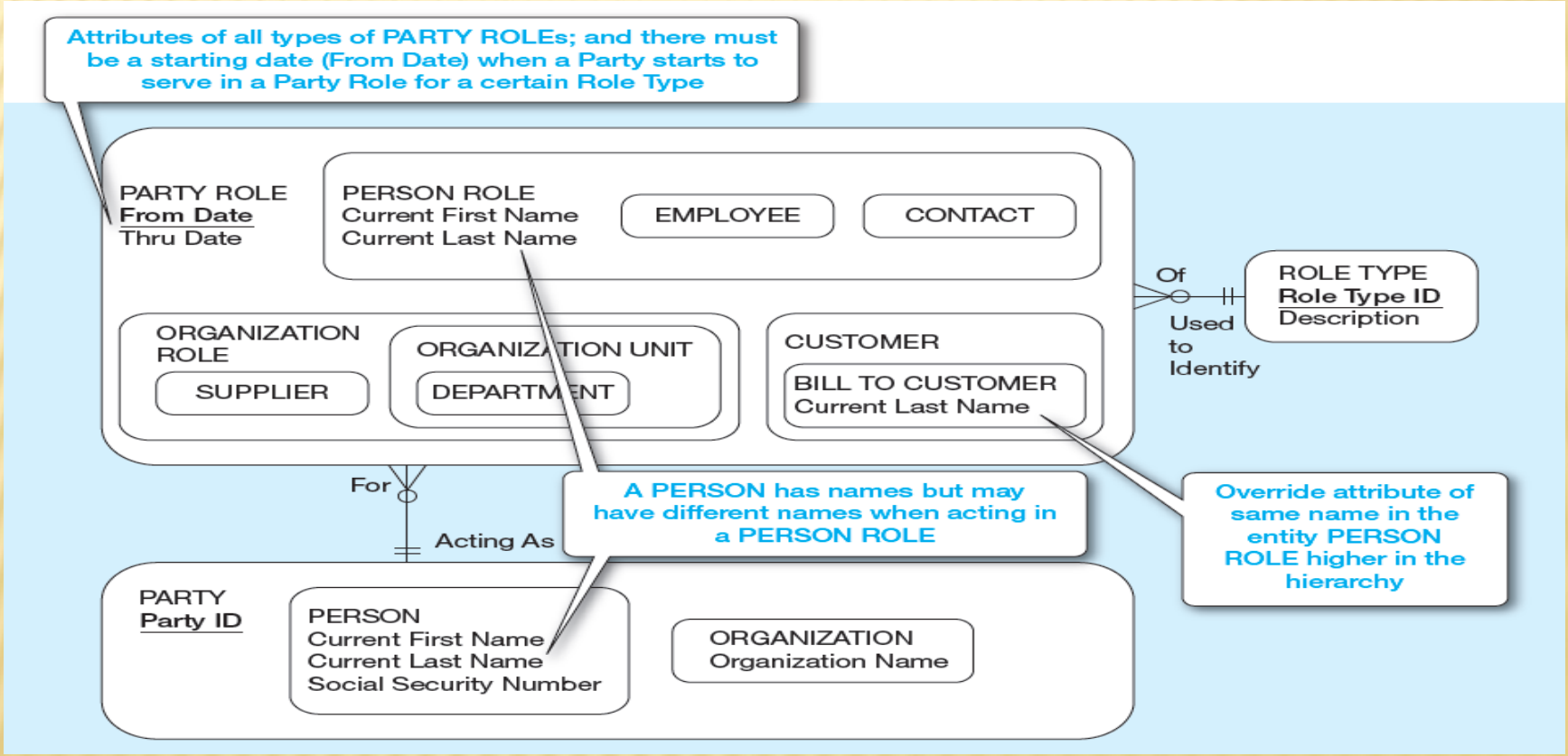


FIGURE 3-15 PARTY, PARTY ROLE, AND ROLE TYPE IN A UNIVERSAL DATA MODEL (2 OF 2)

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b) PARTY supertype/subtype hierarchy

